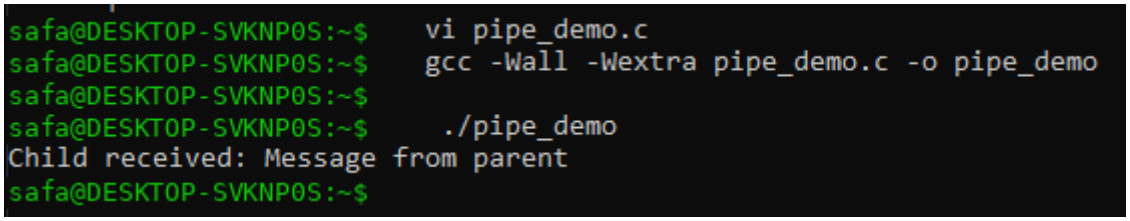
Experiment 17: Inter-Process Communication (Pipes, FIFOs & Message Queues)

Source file: pipe_demo.c

Source Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <unistd.h>
int main(void) {
   int fd[2];
   if (pipe(fd) == -1) { perror("pipe"); return EXIT_FAILURE; }
   pid_t pid = fork();
   if (pid < 0) { perror("fork"); return EXIT_FAILURE; }
   if (pid == 0) {
      close(fd[1]);
      char buffer[100];
      ssize_t n = read(fd[0], buffer, sizeof buffer - 1);
      if (n < 0) { perror("read"); _exit(1); }
      buffer[n] = '\\0';
      printf("Child received: %s\\n", buffer);
      close(fd[0]);
      _exit(0);
   }
   close(fd[0]);
   const char *message = "Message from parent";
   if (write(fd[1], message, strlen(message)) < 0) perror("write");
   close(fd[1]);
   waitpid(pid, NULL, 0);
   return 0;
}
```

Output :